

Chapter 8 — Metadata and data dictionaries

Knowing that documentation matters is not enough, teams need concrete components

Learning objectives

- By the end of this part

Documentation as a user manual

- Dataset documentation functions as a blueprint or user manual
- It describes what the dataset represents, how it was collected or generated
- Reducing misuse in collaborative settings

Core metadata fields

- Metadata is data about data
- Data sources for authenticity and quality assessment
- Version information adds release numbers and change notes so users can track how the

Example 8.5 — Air-Quality Metadata Fields

- Example 8.5 — hands-on module
- Example 8.5 lists metadata for an air-quality monitoring corpus
- Those fields tell a user whether the release fits a modeling or reporting task
- Explore the chapter example module
- View files: ``modules/chapter8/example5/``

What a data dictionary records

- A data dictionary explains each variable
- Consistent entries prevent misinterpretation of columns and reduce errors when analysts
- Completeness of the dictionary is as important as completeness of the data rows

Example 8.6 — Age Variable Description

- Example 8.6 — hands-on module
- Example 8.6 shows a concise dictionary description
- Short, unambiguous phrasing beats vague labels that leave units or populations unspecified
- Explore the chapter example module
- View files: ``modules/chapter8/example6/``

Example 8.8 — Purchase Amount Data-Dictionary Entry

- Example 8.8 — hands-on module
- Example 8.8 records a structured entry for purchase amount
- Explore the chapter example module
- View files: ``modules/chapter8/example8/``

Example 8.8 — listing

```
[
  {
    "Variable Name": "purchase_amount",
    "Data Type": "float",
    "Description": "Total amount spent by the customer in the month",
    "Allowed Values": "Any positive number",
    "Units": "USD"
  }
]
```

Takeaways

- Metadata situates the whole dataset; the data dictionary situates each variable
- Creators, dates, sources, and versions answer who and when
- Structured dictionary entries reduce ambiguity for both humans and automated consumers

Next

- Complete the quiz for this part
- Annotations that capture sampling limits and known faults